



LAB 02

PH142 Summer 2026

Announcements

- **Lab02:** due 7/8 at 11:59pm
- **Quiz03:** due 7/8 at 11:59pm
 - Daily quizzes are available all day (24h)
 - Once opened, you have 30 minutes to complete the quiz



Week 2 Lecture Review

Exploring Data

- Get to know your data before analyzing!
- Key functions:
 - `head(data)`: view the first few rows
 - `dim(data)`: view the number of rows and columns
 - `names(data)`: view column names
 - `str(data)`: view the types of variables

Week 2 Lecture Review

dplyr Review (P1)

Function	Purpose	Input	Output	Notes
<code>rename()</code>	Rename variables	<code>rename(old_dataset, new_name = old_name)</code>	Renamed dataset	Can do multiple variables at one!
<code>select()</code>	Select a subset of variables	<code>select(dataset, column1, column2)</code>	Dataset with specific columns	Can remove variables with -
<code>arrange()</code>	Sort by a variable	<code>arrange(dataset, column)</code>	Dataset arranged by specific columns	Can sort by multiple variables, default is ascending order, descending order with -

Week 2 Lecture Review

dplyr Review (P2)

Function	Purpose	Input	Output	Notes
<code>filter()</code>	Select a subset of rows	<code>filter(dataset, condition)</code>	Dataset with rows based on conditions	<code>==, <, >, <=, >=, !=</code> Or <code>()</code> , <code>and()</code> , <code>&</code>
<code>mutate()</code>	Add new variables to a dataset	<code>mutate(dataset, new_column = data)</code>	Dataset with new variable	You can call existing variables in the dataset to create your new variable
<code>group_by</code>	Group data by a categorical variable	<code>group_by(column)</code>	Dataset grouped by variable	Use with <code>summarize()</code> !
<code>summarize()</code>	Summarize a statistic	<code>summarize(statistic_name = calculation)</code>	Tibble with grouped data and statistic for each group	Use with <code>group_by()</code> !

Week 2 Lecture Review

Pipe Operators

Key operator: Pipe (`%>%`)

- Connects functions together so you can do more than one in a single code chunk
- Shorthands your code by only calling the dataset once
- Read as “and then”... R will read your functions from left to right

Template: `dataset %>% function1( ) %>% function2( )...`

Week 2 Lecture Review

ggplot

Template: `ggplot(data= dataset, aes(x=var1, y=var2)) + geom_point( )`

- `geom_line( )`
- `geom_histogram( )`
- `geom_bar( )`

Other Changes:

- `+ labs(title=" ", y=" ", x=" ")`
- `aes(col= variable1, lty=variable2)`

Week 2 Lecture Review

Types of Variables

Categorical variables: variables that have grouping levels

- Nominal variables: have no underlying order or rank
- Ordinal variables: can be ordered or ranked

Quantitative variable: continuous, numeric variables that that you can perform mathematical operations on

- Discrete variables: can be counted
- Continuous variables: can be measured precisely

Week 2 Lecture Review

Visualizing Distributions

Bar Charts (categorical variables):

- Y-axis: a count/percentage of each category

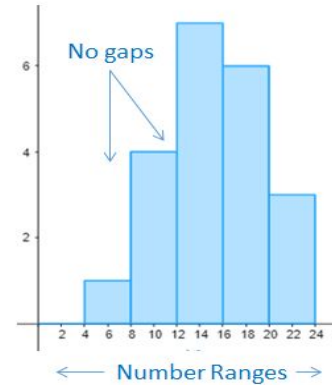


← Categories →

Bar Chart

Histograms (quantitative variables):

- Distribution of values across bins
- Describe shape, center, spread, outliers



← Number Ranges →

Histogram



LAB 02 Walkthrough

Lab Submission

- Follow the directions on the LAB02 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**